THE LARGEST OPEN SUBSCHEME ON WHICH TWO SECTIONS OF AN ETALE SHEAF AGREE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the construction of W and the verification of its characterising property are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Sheaf of sets on the small etale site and the representable presheaf of an open subscheme. Same chapter, the sheaf condition for sets on S_etale and the sheaf represented by an open subscheme. Supplies the presheaf used to produce the open W.
- Sheafification and the universal property of the associated sheaf. Same chapter, the universal property of sheafification, applied to the presheaf of morphisms along which the two sections agree. Supplies the identification of W with the required locus.

Source reference. Original lemma 04HV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3923–3923. Navigation pointer invoked at proof line 4950; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03Q1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4649–4649. Navigation pointer invoked at proof line 4955; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a sheaf of sets on $S_{étale}$. Let $s, t \in \mathcal{F}(S)$. Then there exists an open $W \subset S$ characterized by the following property: A morphism $f : T \to S$ factors through W if and only if $s|_T = t|_T$ (restriction is pullback by f_{small}).*

Proof. Consider the presheaf which assigns to $U \in \mathrm{Ob}(S_{étale})$ the empty set if $s|_U \neq t|_U$ and a singleton else. It is clear that this is a subsheaf of the final object of $Sh(S_{étale})$. By Lemma 04HV we find an open $W \subset S$ representing this presheaf. For a geometric point $\overline{x}$ of S we see that $\overline{x} \in W$ if and only if the stalks of s and t at $\overline{x}$ agree. By the description of stalks of pullbacks in Lemma 03Q1 we see that W has the desired property. □